

# Maralbayev Adil

DevOps / DevSecOps Engineer

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Astana, Kazakhstan

## SUMMARY

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DevOps / DevSecOps Engineer with 3+ years of hands-on experience building and operating on-premise production infrastructure — 200+ virtual machines across 15+ hypervisors and 15+ Kubernetes clusters — serving 30+ B2B and B2G systems at 99.9%+ uptime, including government services used Kazakhstan-wide. Specialised in bare-metal and hypervisor-based Kubernetes, Docker Swarm, GitLab CI/CD with embedded security gates, and Infrastructure as Code with Ansible and Terraform. Prepared and passed state information security certification for third-class (K3) government information systems. Strong background in Linux and PostgreSQL hardening, network segmentation and DMZ design, high availability, disaster recovery, and observability. Backend development background in Go, applied daily to automation and to faster root-cause analysis with product teams.

## TECHNICAL SKILLS

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- **Containers & Orchestration:** Kubernetes (kubeadm, HA control plane, cluster scaling), Helm, kubectl, Docker, Docker Swarm, containerd
- **CI/CD & DevSecOps:** GitLab CI/CD, pipeline caching and build optimisation, secret scanning (gitleaks / trufflehog), SAST, container image scanning (Trivy), GitLab secret & environment separation, artifact registries
- **IaC & Automation:** Ansible, Terraform, Helm charts and manifests, Bash, Go, Python, n8n workflow automation, Claude Code / LLM-assisted automation
- **Infrastructure & Virtualisation:** On-premise bare metal, Proxmox VE, XCP-ng / Xen Orchestra, VMware Cloud Director, Linux (Ubuntu, RHEL-based)
- **Networking & Security:** DMZ segmentation, VLANs, firewalling (iptables / nftables / ufw), HAProxy, Nginx, VPN tunnels, SSH & PAM hardening, auditd, CIS-style OS baselines, least-privilege access control, TLS / PKI
- **Databases:** PostgreSQL (Patroni HA, replication, tuning, pgBackRest / WAL archiving, TimescaleDB, role & network hardening)
- **Observability:** Prometheus, Grafana, Alertmanager, Node / Blackbox / cAdvisor / SQL exporters, Elasticsearch, Logstash, Kibana
- **Backup & DR:** Veeam Agent for Linux, Vinchin, Proxmox Backup Server, logical & physical PostgreSQL backups, restore drills

## PROFESSIONAL EXPERIENCE

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- **Hardcode** Astana, Kazakhstan  
*DevOps / DevSecOps Engineer* *July 2023 – Present*
  - Designed and delivered on-premise production infrastructure from bare metal upward — hypervisors, virtual machines, network segmentation, storage, backup and monitoring — growing the estate to **200+ virtual machines across 15+ hypervisors** and supporting **30+ production customer systems** at **99.9%+ uptime**, including state services operated Kazakhstan-wide.
  - Reduced GitLab CI/CD pipeline build time by **approximately 50%** by introducing layered Docker build caching, dependency caches and shared runner strategy, cutting delivery feedback loops for development teams.
  - Embedded security gates directly into CI/CD as mandatory pipeline stages — automated secret detection (passwords, API keys, private keys, tokens), container image vulnerability scanning and configuration linting — preventing credentials and vulnerable images from ever reaching build artifacts or production registries.
  - Prepared and successfully certified **two government projects** against Kazakhstan information security requirements as **third-class (K3) information systems** — the classification applied to state information systems under the Unified Technical Requirements, verified by attestation testing of the State Technical Service (GTS) before a system may be put into commercial operation. Owned the full cycle: gap analysis against the requirements, OS and application hardening, evidence and documentation package, coordination with SOC and auditors during assessment, and remediation of all identified findings.
  - Deployed, configured and operated **15+ highly available Kubernetes clusters on on-premise infrastructure end to end and independently** — each spanning **15–25 nodes** across 2–3 physical servers, with multi-master control plane, external etcd, load balancing on a keepalived VIP, CNI, ingress, persistent storage and HA application topologies — with no managed cloud control plane to fall back on.

- Scaled production Kubernetes clusters horizontally on multiple occasions, adding and safely draining / removing worker nodes with **zero downtime** to match workload growth and to decommission hardware, including certificate rotation and control-plane maintenance.
- Hardened Linux estate to security-baseline standard: SSH hardening (key-only auth, restricted ciphers and access), PAM policies (password complexity, lockout, session limits), auditd rule sets, kernel and sysctl tuning, firewall policy, and least-privilege access control for virtual machines, databases and services — rolled out consistently across all stands via Ansible.
- Designed network segmentation and **DMZ zoning** separating public-facing entry points from application and database tiers, with controlled ingress/egress, reverse proxies and VPN-only administrative access, meeting regulator requirements for state systems.
- Administered PostgreSQL in production: HA clustering and replication, security configuration (network ACLs, **pg\_hba** policy, role separation, TLS, encrypted credentials) and performance tuning (memory, checkpoint, autovacuum and query optimisation) — reducing slow query load and stabilising high-traffic production databases.
- Built enterprise backup and disaster recovery: Veeam, Vinchin and Proxmox Backup Server for VM backup, replication and restore, plus PostgreSQL logical and physical backups with WAL archiving and point-in-time recovery. Automated scheduling, monitored backup health and ran regular restore drills with a **100% successful restore record** — proving recoverability rather than assuming it.
- Implemented full-stack monitoring and observability across all projects — Prometheus, Grafana, Alertmanager with Node, Blackbox, cAdvisor and SQL exporters, centralised ELK logging, and automated alert routing to email and messengers via n8n — turning silent failures into actionable alerts and cutting incident response time by **40–50%**.
- Standardised infrastructure with Infrastructure as Code: Ansible roles and playbooks and Terraform provisioning for virtual machines, applications, backups and monitoring — reducing manual operational work by **approximately 70%**, eliminating configuration drift and making new environments reproducible instead of hand-built.
- Automated recurring operational work with **Go, Bash and Python** tooling — health checks, backup verification, log rotation, certificate expiry monitoring and bulk fleet operations — removing repetitive manual toil from the team's day.
- Adopted **Claude Code and LLM-assisted engineering** as a working part of the delivery process for automation scripting, IaC authoring, log and incident analysis, and documentation, and introduced AI-assisted diagnostic workflows to colleagues — measurably improving team productivity and response time.
- Provided continuous production support and on-call incident response for 30+ customer systems, sustaining **99.9%+ uptime**: debugging, root cause analysis, system recovery and post-incident follow-up across the full stack, from hypervisor and network through Kubernetes to application and database layers.

#### • **Hardcode**

Astana, Kazakhstan

#### • *Backend Developer (Go)*

2022 – July 2023

- Developed backend services in Go, gaining a working understanding of web application architecture, request lifecycle and common failure modes — applied since as a DevOps engineer to diagnose application-level incidents quickly and collaborate effectively with backend and frontend teams.

## KEY PROJECTS

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- **Government Information Security Certification (K3):** Led infrastructure preparation for two state systems through Kazakhstan's information security attestation as third-class information systems — hardening, access control, documentation and regulator coordination — both passed.
- **On-Premise HA Kubernetes Platform:** Built and operated 15+ multi-master Kubernetes clusters (15–25 nodes each) on company hardware, including HA control plane, load balancing, ingress, persistent storage, GitLab CI/CD delivery and Helm-based release management.
- **Secure CI/CD Pipeline Programme:** Rebuilt GitLab pipelines for Docker Swarm and Kubernetes with build caching (~50% faster builds), mandatory secret-scanning and image-scanning stages, Ansible-driven deployment and strict environment separation.

## EDUCATION

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#### • **Astana IT University (AITU)**

Astana, Kazakhstan

#### • *Bachelor's in Cybersecurity; GPA: 3.5 / 4.0*

Sep. 2021 – June 2024